

SQL Fundamentals: SELECT & Filtering

BrightLearn Basic SQL

Exercise 01 - SQL FOUNDATIONS

01. Selecting and Sorting

(01) SELECT *

FROM employees;

employees

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-10	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

(02) SELECT DISTINCT department

FROM employees;

department
IT
HR
Finance
Marketing

(03) SELECT first_name,
last_name

FROM employees

ORDER BY salary DESC;

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

```
(04) SELECT id,
        first_name,
        last_name,
        salary
FROM employees
ORDER BY salary
LIMIT 3;
```

id	first_name	last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

02. Filtering with WHERE

```
(05) SELECT id,
        first_name,
        last_name,
        department
FROM employees
WHERE department = 'IT';
```

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

(06) SELECT id,
first_name,
last_name,
department,
salary
FROM employees
WHERE department = 'Finance' AND salary > 60000;

id	first_name	last_name	department	Salary
3	Cathy	Zulu	Finance	650 000

(07) SELECT id,
first_name,
last_name,
department
WHERE department = 'HR'
OR department = 'Marketing';

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

(08) SELECT id,
first_name,
last_name,
department
WHERE NOT department = 'IT';

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

(09) SELECT id,
first-name,
last-name,
department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

(10) SELECT id,
first-name,
last-name,
department,
salary,
city
FROM employees
WHERE department = 'IT'
AND salary > 65000
AND city = 'Johannesburg';

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg